

# Aggregate error structure — all 35 models that learned the task

summed over 35 models × 90 annotations · macro AUC ≥ 0.9 · percentages are row-normalised

|              |            |                                           |                                           |                                           |                                           |
|--------------|------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|-------------------------------------------|
| ground truth | thumbout   | <div><div>784</div><div>74.7%</div></div> | <div><div>2</div><div>0.2%</div></div>    | <div><div>87</div><div>8.3%</div></div>   | <div><div>177</div><div>16.9%</div></div> |
|              | openhand   | <div><div>0</div><div>0.0%</div></div>    | <div><div>988</div><div>94.1%</div></div> | <div><div>0</div><div>0.0%</div></div>    | <div><div>62</div><div>5.9%</div></div>   |
|              | closedhand | <div><div>89</div><div>8.5%</div></div>   | <div><div>1</div><div>0.1%</div></div>    | <div><div>784</div><div>74.7%</div></div> | <div><div>176</div><div>16.8%</div></div> |
|              | background | <div><div>124</div><div>38.8%</div></div> | <div><div>114</div><div>35.6%</div></div> | <div><div>82</div><div>25.6%</div></div>  | <div><div>—</div><div></div></div>        |
|              |            | thumbout                                  | openhand                                  | closedhand                                | background                                |
|              |            | predicted                                 |                                           |                                           |                                           |

176 of 179 class confusions (98.3%) are thumbout ↔ closedhand. Only 3 involve openhand.

mean per-class AUC: thumbout 0.9497 openhand 0.9979 closedhand 0.9550

The 35 models are not independent trials — they are scored on the same 90 annotations, so an intrinsically ambiguous hand is counted repeatedly. The matrix shows the shape of the error reliably; read its magnitude as "how often across configurations", not as 3,150 independent observations.